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To the Recruitment Team

Robert Bosch Manufacturing Solutions GmbH

Stuttgart, Germany

May 7, 2026

Subject: Application for PhD Position: AI-based Learning Strategies for Smart Manufacturing (REF283850L)

Dear Members of the Selection Committee,

I am writing to express my strong interest in the PhD position within the European HORIZON project at Robert Bosch Manufacturing Solutions. As a final-year AI and Data Science Engineer from ESI-SBA, specializing in high-fidelity Digital Twins and sequential modeling, I am eager to apply my expertise in **General Purpose AI (GPAI)** and **System 1/2** architectures to the field of robotic machine loading in Smart Manufacturing.

My Master's thesis at **Sonatrach** directly addresses the core challenges described in your offer. I developed a hardware-software co-designed Digital Twin for the LM2500 gas turbine, where I bridged the gap between fast perception and deep reasoning:

- **System 1/2 Integration:** I utilized TCNs for rapid anomaly detection (System 1) and developed a **Mamba-SSM/xLSTM** surrogate model (System 2) that acts as a "What-If" simulator, predicting physical trajectories under counterfactual scenarios.
- **World Models & Simulation:** By integrating physics-informed constraints (PINNs), I created a robust world model that adheres to thermodynamic laws, preventing the "hallucinations" common in pure data-driven models.
- **Real-Time Edge Performance:** I optimized these models for edge deployment using C++ JIT compilation, achieving a record-breaking inference latency of **0.04ms**, a critical requirement for high-frequency robotic control loops.

The HORIZON project's focus on **General Purpose AI** for (humanoid) robots aligns perfectly with my research trajectory. My experience with **Explainable AI (XAI)**—specifically using gradient saliency heatmaps to diagnose industrial failures—demonstrates my commitment to the principles of Trustworthy AI. Furthermore, my proficiency in Python, PyTorch, and ROS, combined with my ability to work in dynamic, uncertain environments, ensures I can contribute effectively to your international research consortium.

I am particularly motivated by the opportunity to validate these GPAI tools at the **ARENA2036** in Stuttgart. I am confident that my background in hybrid modeling and real-time inference will provide a solid foundation for advancing the state-of-the-art in smart manufacturing.

Thank you for your time and consideration. I look forward to the possibility of discussing how my technical skills and research background can contribute to the success of Bosch's robotic initiatives.

Sincerely,

Ismail AISSAOUI